

Cohort Analysis using Python

Cohort analysis is a data analysis technique used to gain insights into the behavior and characteristics of specific groups of users or customers over time. If you want to learn how to perform cohort analysis, this article is for you. In this article, I'll take you through the task of cohort analysis using Python.

Cohort analysis is valuable for businesses as it allows them to understand user behaviour in a more granular and actionable way. Below is the process we can follow for the task of Cohort Analysis:

1. The first step is to define the cohorts based on a specific characteristic or event. For example, in an e-commerce platform, cohorts could be defined based on the month of a user's first purchase.
2. Gather relevant data for analysis.
3. Determine the time intervals you want to analyze.
4. Group users into cohorts based on the defined characteristic or event.
5. Choose the key performance metrics you want to analyze.
6. Calculate the chosen metrics for each cohort over the specified time periods.
7. Create visualizations to present your findings effectively.

A dataset for Cohort Analysis typically includes user or customer data, such as registration date, purchase history, engagement metrics, or any other data points relevant to the analysis. I found an ideal dataset for this task.

The descriptive statistics provide the following insights:

- **New Users:** The average number of new users is around 3,418 with a standard deviation of approximately 677. The minimum and maximum new users recorded are 1,929 and 4,790, respectively.
- **Returning Users:** On average, there are about 1,353 returning users, with a standard deviation of around 247. The minimum and maximum are 784 and 1,766, respectively.
- **Duration Day 1:** The average duration on the first day is about 208 seconds with a considerable spread (standard deviation is around 65).
- **Duration Day 7:** The average 7-day duration is lower, around 136 seconds, with a larger standard deviation of about 97. The range is from 0 to 304.

For the task of Cohort Analysis, we'll group the data by the week of the year to create cohorts. Then, for each cohort (week), we'll calculate the average number of new and returning users, as well as the average of Duration Day 1 and Duration Day 7. Let's start by grouping the data by week and calculating the necessary average

We can see that the number of new users and returning users fluctuates from week to week. Notably, there was a significant increase in both new and returning users in Week 47. The average duration of user engagement on Day 1 and Day 7 varies across the weeks. The durations do not follow a consistent pattern about the number of new or returning users, suggesting that other factors might be influencing user engagement.

Summary

Cohort Analysis is a data analysis technique used to gain insights into the behaviour and characteristics of specific groups of users or customers over time. It is valuable for businesses as it allows them to understand user behaviour in a more granular and actionable way. I hope you liked this article on Cohort Analysis using Python. Feel free to ask valuable questions in the comments section below.